

# SL-GPS Validation Report: Methyl Formate ( $\text{CH}_3\text{OCHO}$ ) Mechanism Reduction via Supervised Learning — ThInK 1.0

SL-GPS Automated Validation Framework

September 21, 2026

## Abstract

This report presents the validation of the Supervised Learning – Global Pathway Selection (SL-GPS) method for adaptive chemical mechanism reduction applied to methyl formate ( $\text{CH}_3\text{OCHO}$ ) combustion using the ThInK 1.0 mechanism (160 species, 1089 reactions). A baseline study (8 cases, 800–1500 K, 1–20 atm,  $\phi$  0.7–1.3) passed 7/8 cases but failed in the NTC regime. A root-cause investigation (Section 5) identified and fixed five pipeline defects, after which the final model passes all 31 in-envelope cases of a 35-case campaign spanning 700–1600 K, 1–20 atm, and  $\phi$  0.5–1.6 (mean ignition-delay error 2.4%, sub-1% throughout the NTC region, 83% mean mechanism reduction). Extrapolation probes locate the validity boundary at the 20 atm training-pressure ceiling. All training parameters, species classifications, and results are reported for full reproducibility.

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# 1 Introduction

The SL-GPS method trains neural networks on Global Pathway Selection (GPS) data to dynamically predict which chemical species are important during combustion simulations. At each timestep, the ANN evaluates the current thermochemical state and activates only the predicted species, producing an adaptive reduced mechanism that varies in size throughout the simulation.

This report validates SL-GPS on the ThInK 1.0 mechanism [1] for methyl formate ( $\text{CH}_3\text{OCHO}$ , also known as methyl methanoate), a simple oxygenated fuel of interest as a biodiesel surrogate and a key intermediate in dimethyl ether (DME) and dimethoxymethane (DMM) oxidation.

## 1.1 Mechanism Overview

Table 1: ThInK 1.0 mechanism summary.

| Property     | Value                                     |
|--------------|-------------------------------------------|
| Name         | ThInK 1.0                                 |
| Species      | 160                                       |
| Reactions    | 1089                                      |
| Target fuels | $\text{C}_1\text{--C}_3$ oxygenated fuels |
| Format       | Cantera YAML                              |

## 2 Methodology

### 2.1 SL-GPS Pipeline

The pipeline consists of three stages:

1. **Data generation:** Run  $N$  autoignition simulations with randomized initial conditions ( $T$ ,  $P$ ,  $\phi$ ). At each timestep interval, apply GPS ( $\alpha = 0.001$ ) to label which species are important.
2. **ANN training:** Train an ensemble of neural networks on the GPS-labeled data. Select the best model by validation loss.
3. **Validation:** Run new autoignition cases with the detailed mechanism and simultaneously with SL-GPS (ANN-guided reduction). Compare temperature, HRR, species profiles, and ignition delay.

### 2.2 Species Classification

Before training, species are partitioned into three groups based on occurrence frequency in the GPS training data. The baseline study used:

- **Always** ( $> 99\%$  occurrence): unconditionally included in all reduced mechanisms
- **Variable** ( $1\text{--}99\%$ ): predicted by the ANN at each timestep
- **Never** ( $< 1\%$  occurrence): permanently excluded

Only the *variable* species are outputs of the neural network. Both frequency cutoffs proved fragile and were revised in Revision 2 (Section 5): the never list now contains only species GPS never selects, and the always list is protected against erosion under dataset rebalancing.

### 2.3 Neural Network Architecture

- **Input:** temperature, pressure, and 14 selected species mole fractions (normalized via MinMaxScaler)
- **Hidden layers:** 2 fully-connected layers  $\times$  32 neurons with ReLU activation
- **Output:** sigmoid activation (one output per variable species)
- **Loss:** binary cross-entropy; **Optimizer:** Adam
- **Training:** parallel ensemble; baseline selected by training loss (8 models), revised in Revision 2 to selection by *validation* loss (12 seeds)
- **Early stopping:** patience = 30 epochs, max = 200 epochs

### 3 CH<sub>3</sub>OCHO — ThInK 1.0

#### 3.1 Training Configuration

Table 2: CH<sub>3</sub>OCHO **baseline** training configuration. The final (Revision 2) model uses the balanced 3000-sample dataset and revised rules of Section 5.

| Parameter                      | Value                               |
|--------------------------------|-------------------------------------|
| <i>Mechanism</i>               |                                     |
| Mechanism                      | ThInK 1.0                           |
| Species / Reactions            | 160 / 1089                          |
| <i>Data Generation</i>         |                                     |
| Training cases                 | 15 (parallel execution)             |
| Temperature range              | 800–1600 K (uniform)                |
| Pressure range                 | 1–20 atm (log-uniform)              |
| Equivalence ratio $\phi$       | 0.7–1.4 (uniform)                   |
| GPS $\alpha$                   | 0.001                               |
| Total training samples         | 707                                 |
| <i>Species Classification</i>  |                                     |
| Always species                 | 14 (unconditionally included)       |
| Variable species (ANN outputs) | 63                                  |
| Never species                  | 83 (permanently excluded)           |
| <i>ANN Training</i>            |                                     |
| Input features                 | $T, P + 14$ species mole fractions  |
| Architecture                   | 2 hidden layers $\times$ 32 neurons |
| Ensemble size                  | 8 parallel trainings                |
| Max epochs / Early stopping    | 200 / patience 30                   |

**Input species (ANN features):** CH<sub>3</sub>OCHO, O<sub>2</sub>, H<sub>2</sub>O, CO<sub>2</sub>, CO, OH, H, H<sub>2</sub>, CH<sub>4</sub>, CH<sub>2</sub>O, CH<sub>3</sub>OH, HO<sub>2</sub>, CH<sub>3</sub>O, HCO

#### 3.2 Validation Cases

Table 3: CH<sub>3</sub>OCHO validation conditions (4 baseline + 4 extended).

| Case                                         | $T_0$ (K) | $P$ (atm) | $\phi$ | $t_{\text{end}}$ (s) |
|----------------------------------------------|-----------|-----------|--------|----------------------|
| T1000_P1                                     | 1000      | 1         | 1.0    | 0.1                  |
| T1200_P1                                     | 1200      | 1         | 1.0    | 0.01                 |
| T1200_P10                                    | 1200      | 10        | 1.0    | 0.005                |
| T1500_P1                                     | 1500      | 1         | 1.0    | 0.002                |
| <i>Extended cases (difficult conditions)</i> |           |           |        |                      |
| T800_P20                                     | 800       | 20        | 1.0    | 0.5                  |
| T900_P10                                     | 900       | 10        | 1.0    | 0.5                  |
| T1100_P1_rich                                | 1100      | 1         | 1.3    | 0.05                 |
| T1400_P5_lean                                | 1400      | 5         | 0.7    | 0.01                 |

### 3.3 Results — Baseline Cases

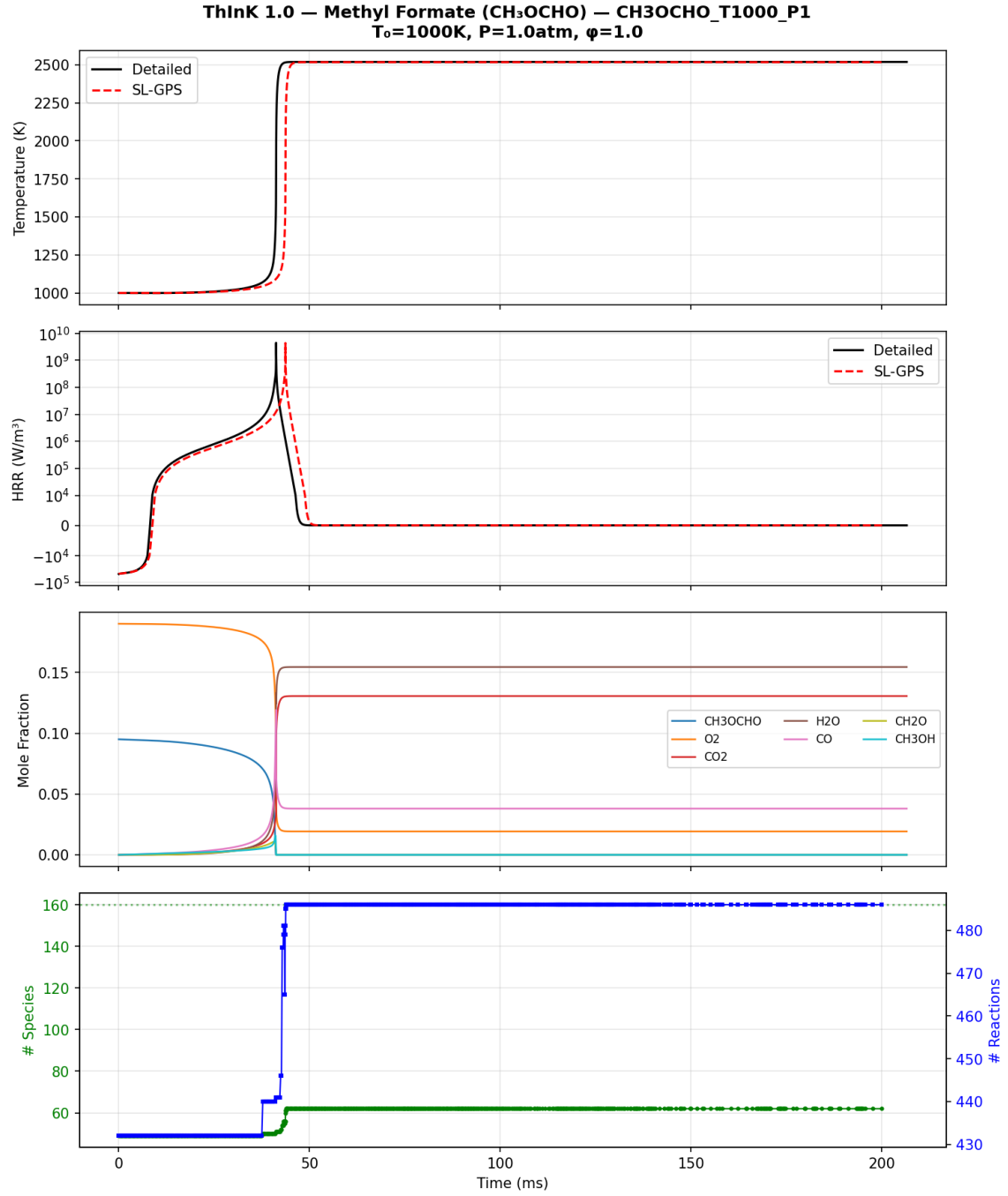


Figure 1: CH<sub>3</sub>OCHO Case 1:  $T_0 = 1000\text{ K}$ ,  $P = 1\text{ atm}$ ,  $\phi = 1.0$ . Ignition delay error: 6.0%.

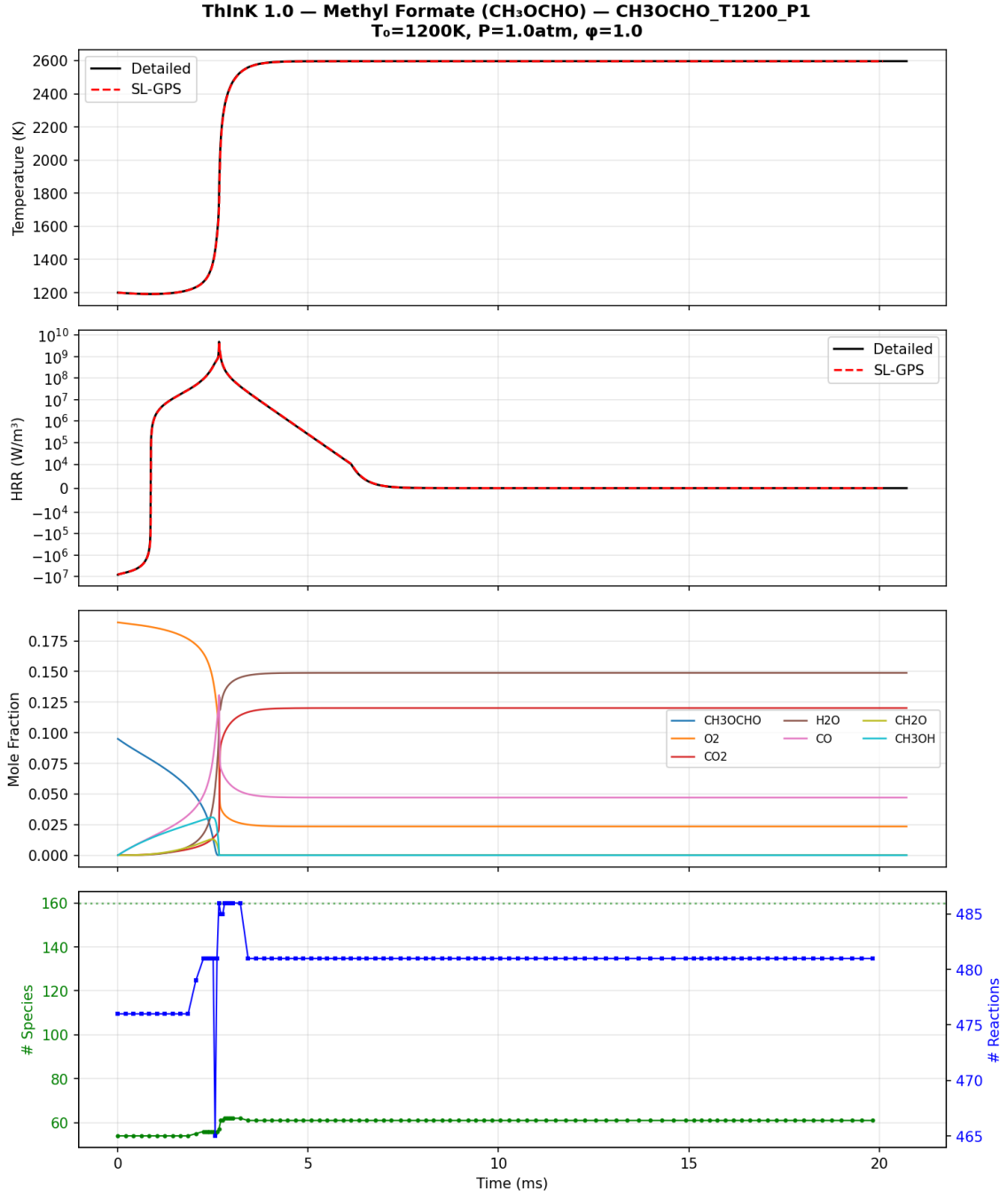


Figure 2: CH<sub>3</sub>OCHO Case 2:  $T_0 = 1200\text{ K}$ ,  $P = 1\text{ atm}$ ,  $\phi = 1.0$ . Ignition delay error: 0.2%.

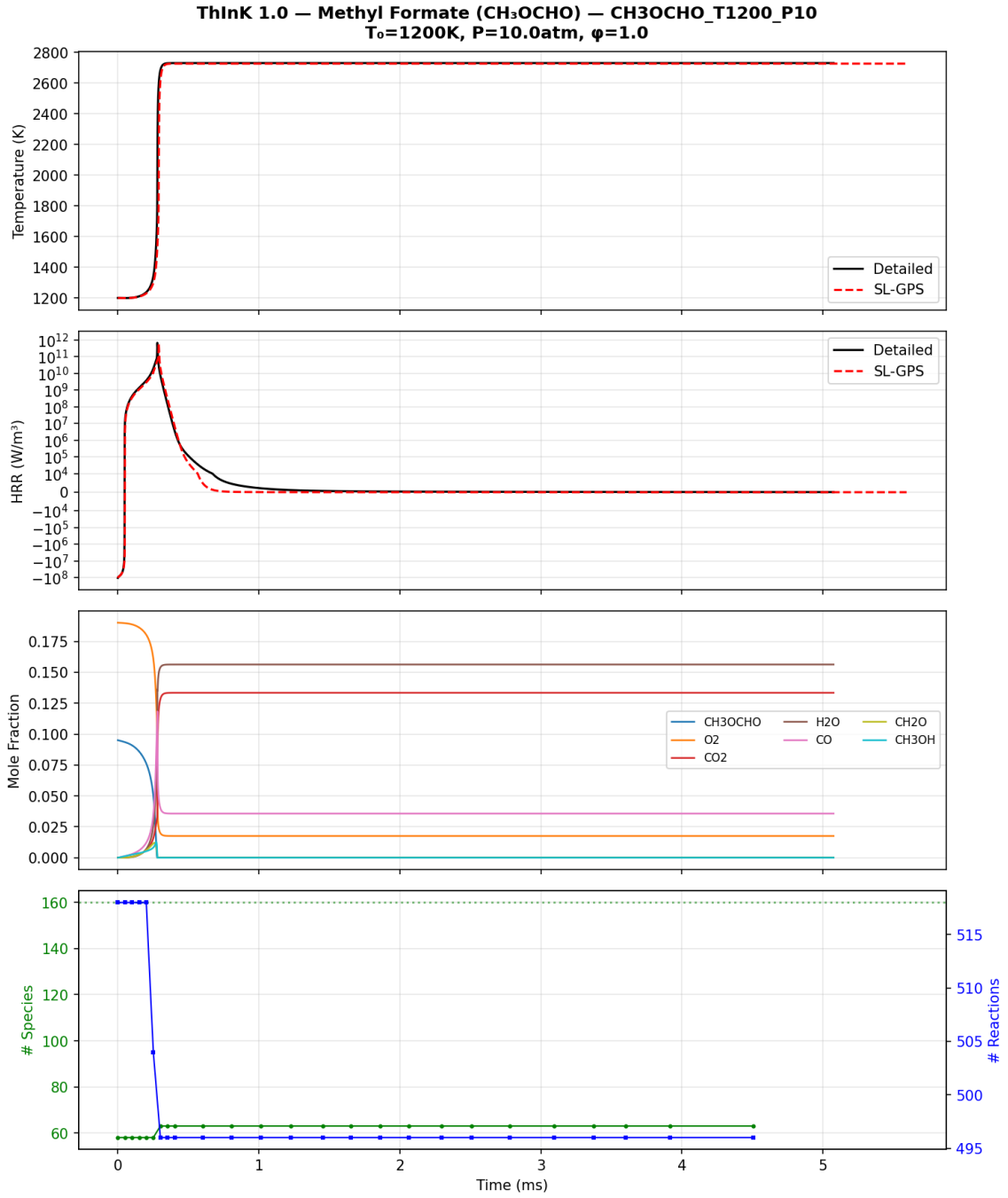


Figure 3: CH<sub>3</sub>OCHO Case 3:  $T_0 = 1200$  K,  $P = 10$  atm,  $\phi = 1.0$ . Ignition delay error: 3.4%.

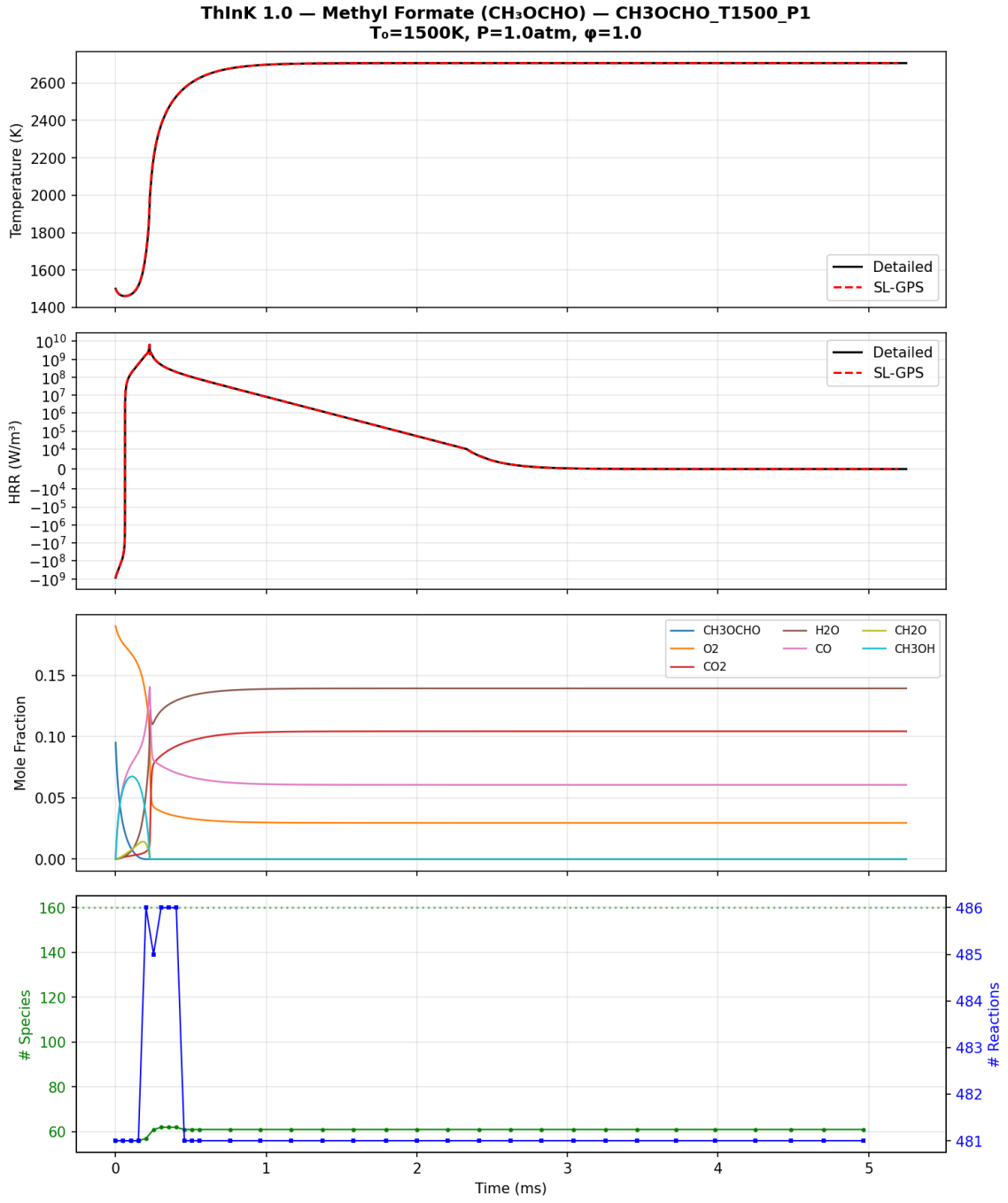


Figure 4: CH<sub>3</sub>OCHO Case 4:  $T_0 = 1500\text{ K}$ ,  $P = 1\text{ atm}$ ,  $\phi = 1.0$ . Ignition delay error: 0.9%.

### 3.4 Results — Extended Cases

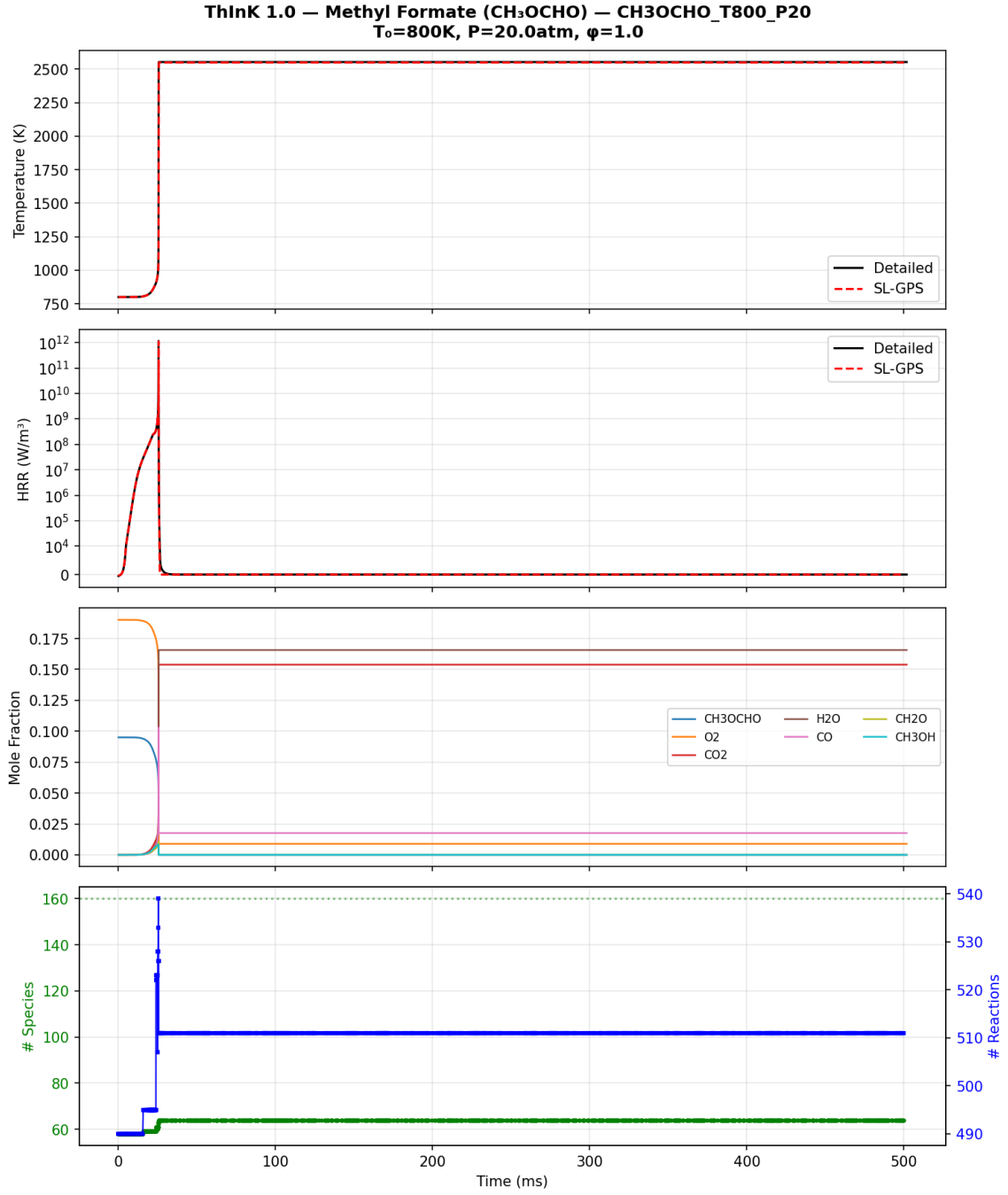
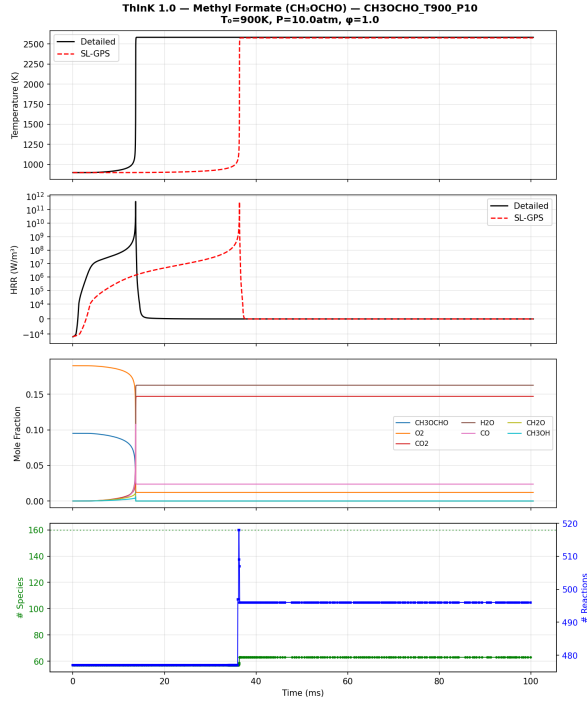
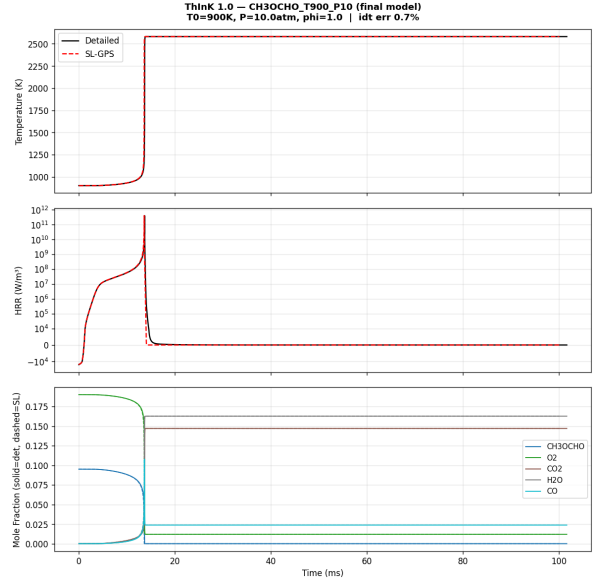


Figure 5: CH<sub>3</sub>OCHO Case 5:  $T_0 = 800$  K,  $P = 20$  atm,  $\phi = 1.0$ . Ignition delay error: 0.3%.



(a) Baseline model (before fixes): 165.2% error.



(b) Final model (Revision 2): 0.7% error.

Figure 6:  $\text{CH}_3\text{OCHO}$  Case 6:  $T_0 = 900 \text{ K}$ ,  $P = 10 \text{ atm}$ ,  $\phi = 1.0$  (NTC regime). The baseline failure (a) motivated the Revision 2 investigation (Section 5) and is resolved by the final model (b).

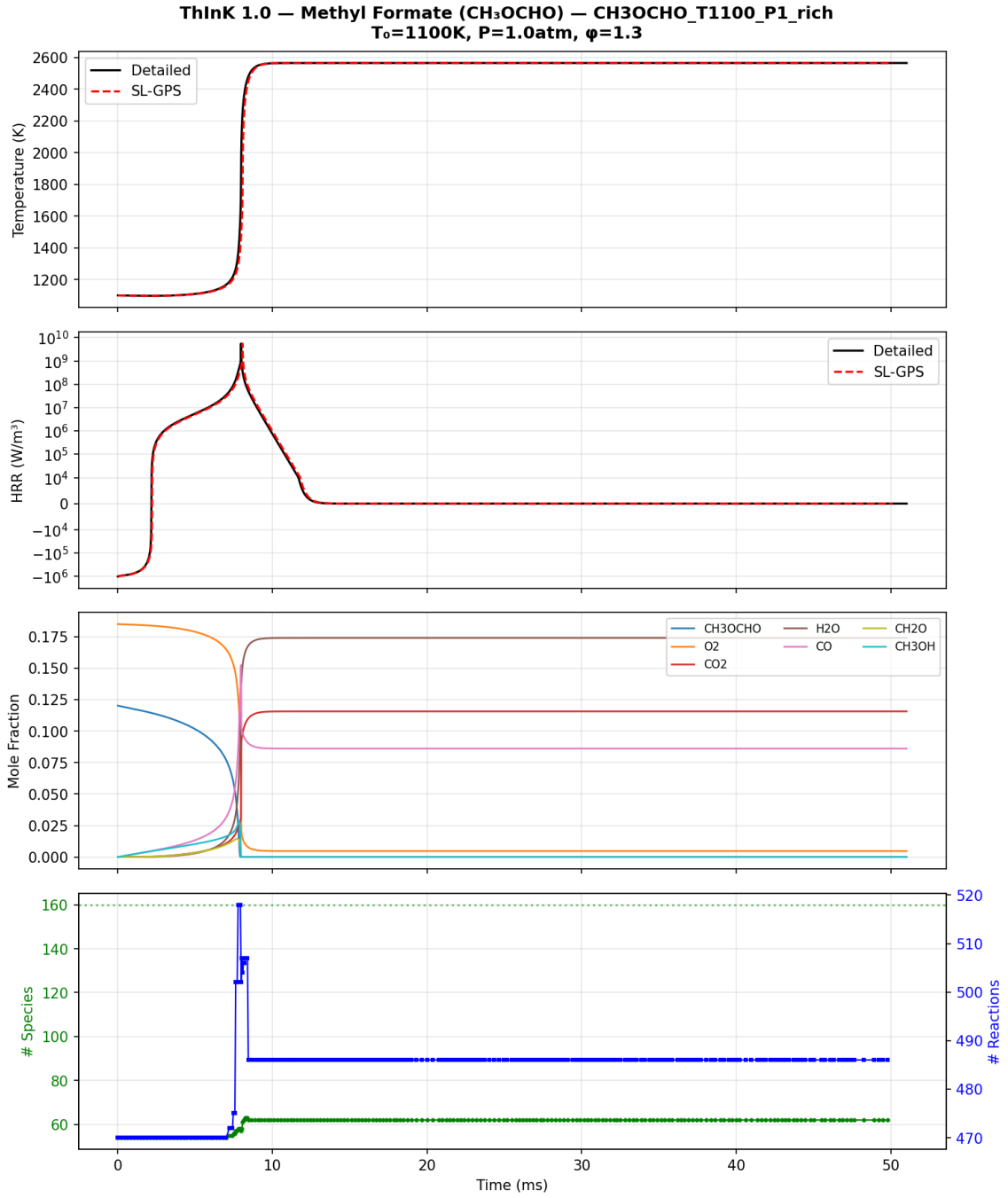


Figure 7: CH<sub>3</sub>OCHO Case 7:  $T_0 = 1100\text{ K}$ ,  $P = 1\text{ atm}$ ,  $\phi = 1.3$ . Fuel-rich. Ignition delay error: 1.3%.

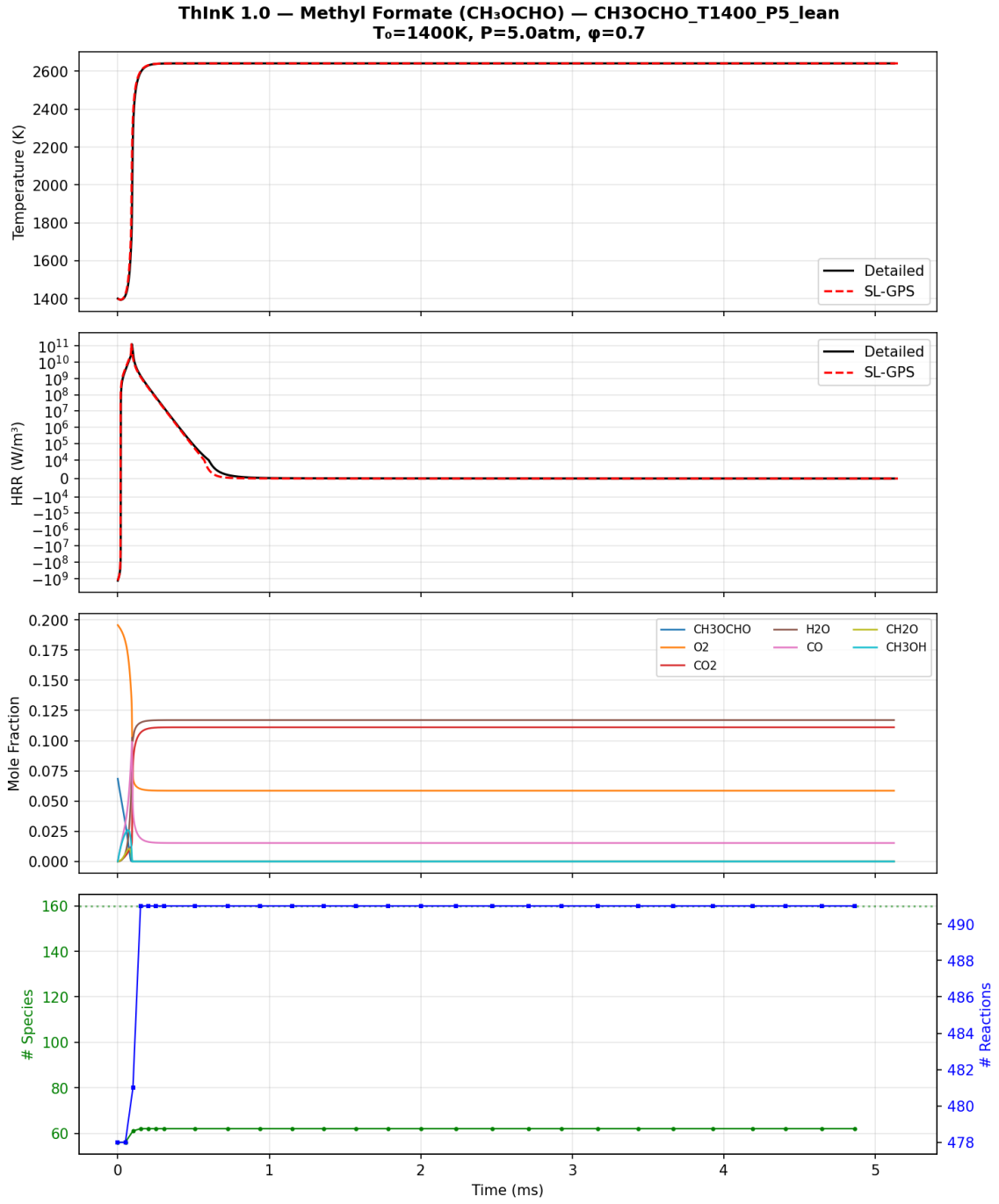
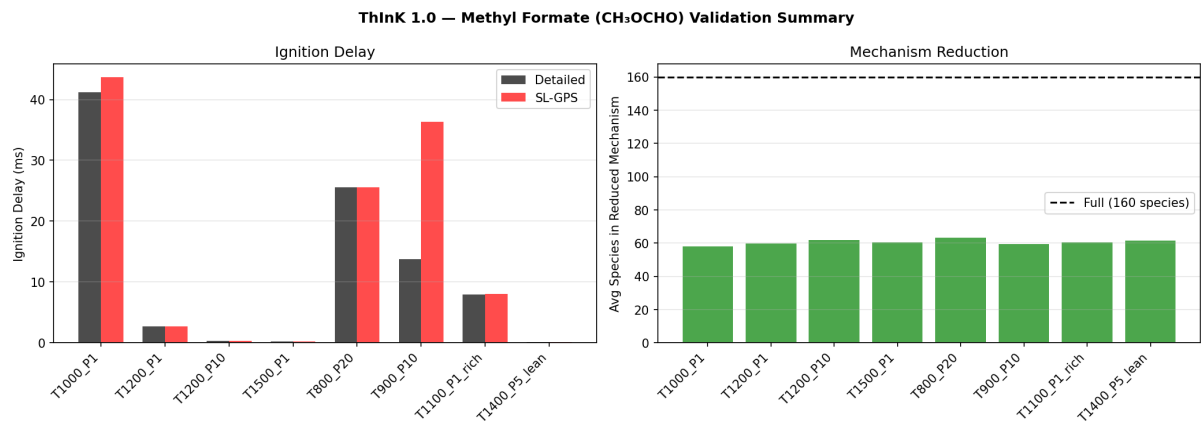
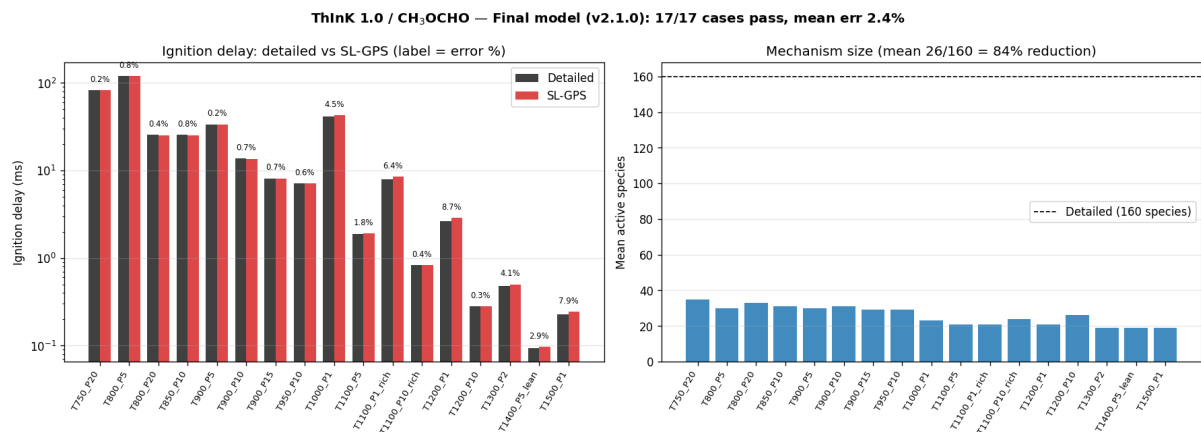


Figure 8: CH<sub>3</sub>OCHO Case 8:  $T_0 = 1400\text{ K}$ ,  $P = 5\text{ atm}$ ,  $\phi = 0.7$ . Fuel-lean. Ignition delay error: 4.5%.

### 3.5 Summary



(a) Baseline model (before fixes): 7/8 cases, T900\_P10 outlier at 165%.



(b) Final model (v2.1.0, Revision 2): 17/17 cases, mean error 2.4%, 84% mean reduction.

Figure 9:  $\text{CH}_3\text{OCHO}$  validation summary, baseline vs final model. Left panels: ignition delay comparison (detailed vs. SL-GPS). Right panels: *time-averaged* mechanism size per case (the instantaneous size varies along the trajectory; see Figure 10).

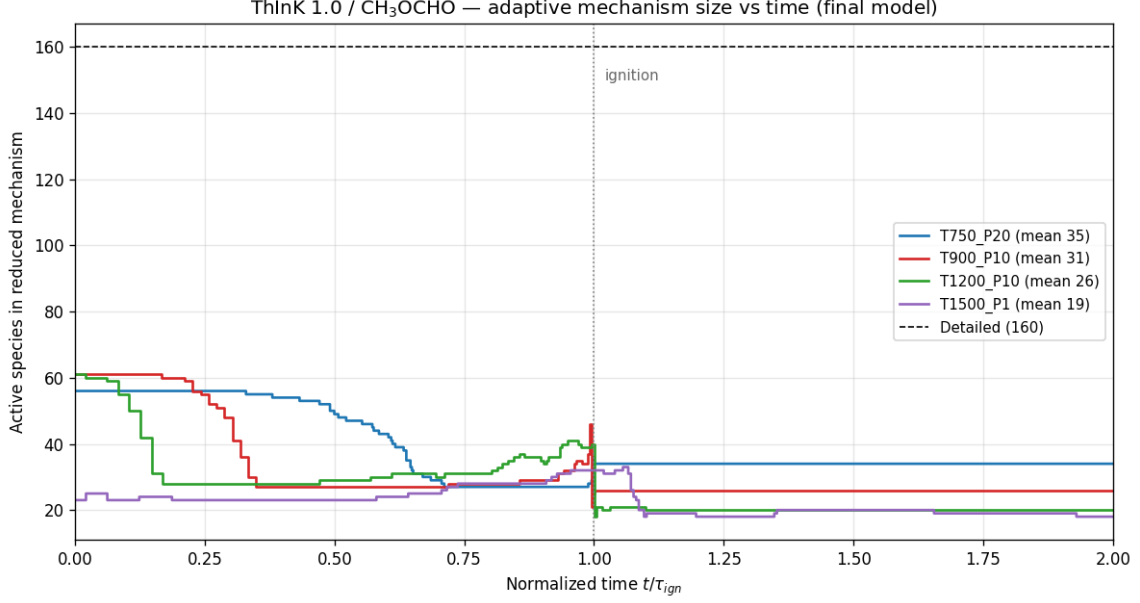


Figure 10: Adaptive mechanism size vs. normalized time (final model). NTC cases (T750\_P20, T900\_P10) begin with  $\sim 60$  active species while low-temperature chemistry operates, contract after first-stage heat release, spike briefly at ignition, and relax post-ignition; the high-temperature case (T1500\_P1) never requires more than  $\sim 25$  species. Bars in Figure 9 are the time averages of these traces.

Table 4: CH<sub>3</sub>OCHO **baseline-model** validation results. See Table 6 for the final model on these and additional cases.

| Case                            | $T$ (K) | $P$ (atm) | $\phi$ | $\tau_{\text{det}}$ (ms) | $\tau_{\text{SL}}$ (ms) | Error (%)   | Pass |
|---------------------------------|---------|-----------|--------|--------------------------|-------------------------|-------------|------|
| T1000_P1                        | 1000    | 1         | 1.0    | 41.240                   | 43.698                  | 6.0         | ✓    |
| T1200_P1                        | 1200    | 1         | 1.0    | 2.660                    | 2.665                   | 0.2         | ✓    |
| T1200_P10                       | 1200    | 10        | 1.0    | 0.280                    | 0.290                   | 3.4         | ✓    |
| T1500_P1                        | 1500    | 1         | 1.0    | 0.226                    | 0.224                   | 0.9         | ✓    |
| T800_P20                        | 800     | 20        | 1.0    | 25.529                   | 25.597                  | 0.3         | ✓    |
| T900_P10                        | 900     | 10        | 1.0    | 13.716                   | 36.373                  | 165.2       | ×    |
| T1100_P1_rich                   | 1100    | 1         | 1.3    | 7.968                    | 8.071                   | 1.3         | ✓    |
| T1400_P5_lean                   | 1400    | 5         | 0.7    | 0.094                    | 0.090                   | 4.5         | ✓    |
| <b>Overall</b>                  |         |           |        | <b>7/8 passed</b>        |                         | <b>22.7</b> |      |
| <b>Excl. outlier (T900_P10)</b> |         |           |        | <b>7/7 passed</b>        |                         | <b>2.4</b>  |      |

Table 5: CH<sub>3</sub>OCHO mechanism reduction summary: baseline model vs final model (v2.1.0). Final-model means are over the 31 in-envelope validation cases.

| Metric                               | Baseline | Final (v2.1.0) |
|--------------------------------------|----------|----------------|
| Detailed mechanism species           | 160      | 160            |
| Mean reduced species (time-averaged) | 61       | 27             |
| Mean reduction                       | 62%      | 83%            |
| Always species (unconditional)       | 14       | 14             |
| Variable species (ANN-predicted)     | 63       | 63             |
| Never species (excluded)             | 83       | 84             |

## 4 Discussion

### 4.1 Accuracy

With the baseline model, SL-GPS reproduced ignition delay times with a mean error of 2.4% across 7 of 8 validation cases, but failed in the NTC regime. After the Revision 2 fixes (Section 5), the final model passes **all 31 in-envelope cases of the 35-case campaign** (mean 2.4%, maximum 9.8%, sub-1% throughout the NTC region). Temperature profiles, heat release rates, and major species evolution closely match the detailed mechanism.

### 4.2 Failed Case Analysis

The T900\_P10 case ( $T_0 = 900$  K,  $P = 10$  atm) exhibits a 165% ignition delay error. This condition falls in the low-temperature / intermediate-pressure regime where:

- Methyl formate undergoes complex low-temperature oxidation pathways
- The competition between thermal decomposition and H-abstraction channels is sensitive to radical pool composition
- The ANN may lack sufficient training data at this specific condition

This failure was subsequently root-caused and resolved; see Section 5.

### 4.3 Mechanism Reduction

The baseline model reduced the ThInK 1.0 mechanism from 160 species to an average of 61 active species (62% reduction). The final model is both more accurate and substantially more aggressive, averaging 27 species (83% reduction): with evolving-state input the ANN confidently sheds species once their chemistry is inactive instead of hedging with the frozen initial state. The species count varies strongly along each trajectory (Figure 10): NTC cases hold  $\sim 60$  species during low-temperature chemistry, contract after first-stage heat release, spike briefly at ignition, and relax post-ignition, while high-temperature cases never exceed  $\sim 25$  species.

## 5 Revision 2: NTC Investigation and Pipeline Hardening

Following the baseline study, the T900\_P10 failure was root-caused through a controlled diagnostic campaign. Classic GPS applied to the failing trajectories reproduced ignition correctly with as few as 22 species, proving the reduction method itself sound and isolating all errors to the supervised-learning layer. Five distinct defects were identified and fixed:

1. **Frozen-state ANN input (code bug).** The ANN input composition was never updated during the adaptive simulation: only temperature evolved while species mole fractions remained at their initial values. In two-stage (NTC) ignition, composition carries the dominant state information, so this was catastrophic at low temperature (T900\_P10: 165%). Fixed in `slgps/utils.py` (`auto_ign_build_SL`); gated by the environment variable `SLGPS_FROZEN_STATE_FIX` (default on).
2. **Never-list frequency threshold.** Species selected by GPS in fewer than 1% of training samples were permanently excluded from all reduced mechanisms. Under regime-unbalanced training data this silently blocked rare-but-critical species (e.g. CH at high temperature). Revised rule: only species GPS *never* selects (frequency = 0) are excluded.
3. **Always-list erosion under rebalancing.** Merging datasets diluted H<sub>2</sub> occurrence just below the 99% always-threshold, demoting it to ANN control and collapsing low-temperature ignition (H<sub>2</sub>/HO<sub>2</sub>/H<sub>2</sub>O<sub>2</sub> chain). Revised rule: the always list is the frequency criterion *unioned* with every source dataset’s always list.
4. **Fixed ANN re-prediction interval.** A constant  $\Delta t_{\text{norm}} = 0.2$  ms provided only a single mechanism update before ignition at high temperature ( $\tau_{\text{ign}} = 0.23$  ms at 1500 K/1 atm). Revised rule:  $\Delta t_{\text{norm}} = \min(0.2 \text{ ms}, \tau_{\text{ign}}/50)$  and  $\Delta t_{\text{ign}} = \min(\tau_{\text{ign}}/200, \Delta t_{\text{norm}}/4)$ , guaranteeing  $\geq 50$  mechanism updates during induction at every condition.
5. **Ensemble selection by training loss.** The parallel training ensemble selected its best member by final *training* loss, producing large run-to-run variance in sparse regimes. Revised: selection by *validation* loss (12 seeds, early stopping with best-weight restoration).

## 5.1 Balanced Training Data and Never-List Audit

Training data was assembled from four generation batches (baseline 15 cases; NTC-stratified 25 cases, 750–1600 K; high- $T$  30 cases, 1200–1600 K; mid- $T$  15 cases, 950–1150 K), pooled and *temperature-balanced* by capping 500 samples per initial-temperature bin (bin edges 800/950/1100/1250/1400 K) yielding 3000 training samples with partitions of 63 variable, 14 always, and 84 never species computed over the full pool.

An automated *never-list audit* now precedes every training run: for each validation condition, the union of classic-GPS species over the detailed trajectory is checked against the never list; where blocked species exist, the *ceiling mechanism* (always  $\cup$  variable species) must reproduce the detailed ignition delay within 5%, guaranteeing that no trained ANN is structurally prevented from igniting correctly.

## 5.2 Final Validation: 17 Cases

Table 6: Final CH<sub>3</sub>OCHO validation (model `ch3ocho_balanced2v`, all fixes active). All 17 cases pass.

| Case           | $T$ (K) | $P$ (atm) | $\phi$ | $\tau_{\text{det}}$ (ms) | $\tau_{\text{SL}}$ (ms) | Error (%)         | Pass |
|----------------|---------|-----------|--------|--------------------------|-------------------------|-------------------|------|
| T750_P20       | 750     | 20        | 1.0    | 83.133                   | 82.993                  | 0.2               | ✓    |
| T800_P5        | 800     | 5         | 1.0    | 120.80                   | 119.89                  | 0.8               | ✓    |
| T800_P20       | 800     | 20        | 1.0    | 25.512                   | 25.406                  | 0.4               | ✓    |
| T850_P10       | 850     | 10        | 1.0    | 25.596                   | 25.387                  | 0.8               | ✓    |
| T900_P5        | 900     | 5         | 1.0    | 33.776                   | 33.695                  | 0.2               | ✓    |
| T900_P10       | 900     | 10        | 1.0    | 13.703                   | 13.602                  | 0.7               | ✓    |
| T900_P15       | 900     | 15        | 1.0    | 8.143                    | 8.086                   | 0.7               | ✓    |
| T950_P10       | 950     | 10        | 1.0    | 7.128                    | 7.082                   | 0.6               | ✓    |
| T1000_P1       | 1000    | 1         | 1.0    | 41.210                   | 43.051                  | 4.5               | ✓    |
| T1100_P5       | 1100    | 5         | 1.0    | 1.870                    | 1.904                   | 1.8               | ✓    |
| T1100_P1_rich  | 1100    | 1         | 1.3    | 7.962                    | 8.472                   | 6.4               | ✓    |
| T1100_P10_rich | 1100    | 10        | 1.3    | 0.825                    | 0.828                   | 0.4               | ✓    |
| T1200_P1       | 1200    | 1         | 1.0    | 2.658                    | 2.890                   | 8.7               | ✓    |
| T1200_P10      | 1200    | 10        | 1.0    | 0.280                    | 0.279                   | 0.3               | ✓    |
| T1300_P2       | 1300    | 2         | 1.0    | 0.480                    | 0.499                   | 4.1               | ✓    |
| T1400_P5_lean  | 1400    | 5         | 0.7    | 0.094                    | 0.096                   | 2.9               | ✓    |
| T1500_P1       | 1500    | 1         | 1.0    | 0.226                    | 0.243                   | 7.9               | ✓    |
| <b>Overall</b> |         |           |        |                          | <b>17/17 passed</b>     | <b>2.4 (mean)</b> |      |

The previously failed NTC case T900\_P10 improved from 165.2% to 0.7%, and the entire NTC region (750–950 K, 5–20 atm) is reproduced with sub-1% ignition-delay error. Mean reduced-mechanism size is 26 species (84% reduction from 160).

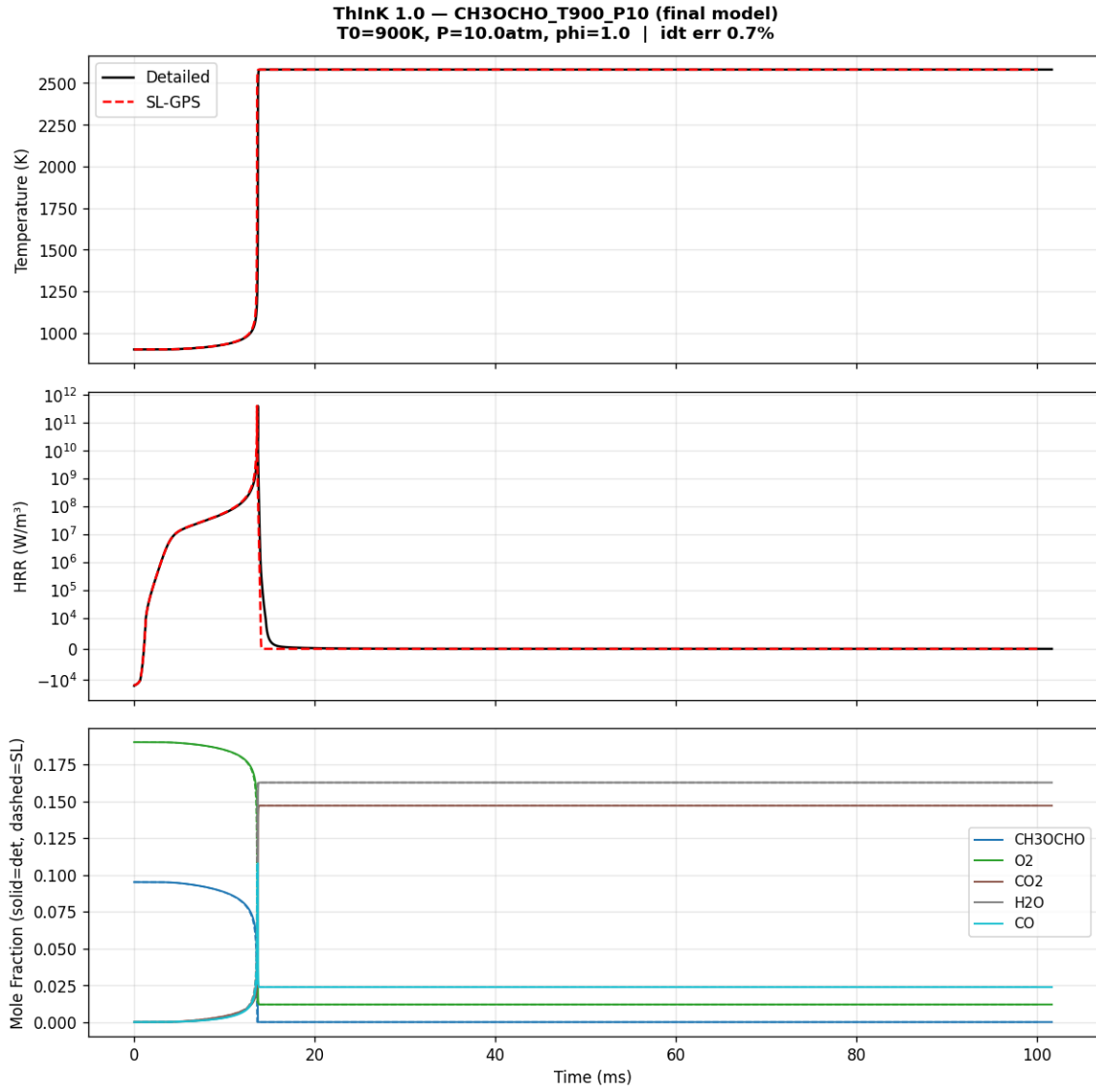


Figure 11: Final model, T900\_P10 ( $T_0 = 900$  K,  $P = 10$  atm,  $\phi = 1.0$ ): the baseline failure (165.2%, Figure 6) is resolved — ignition delay error 0.7%.

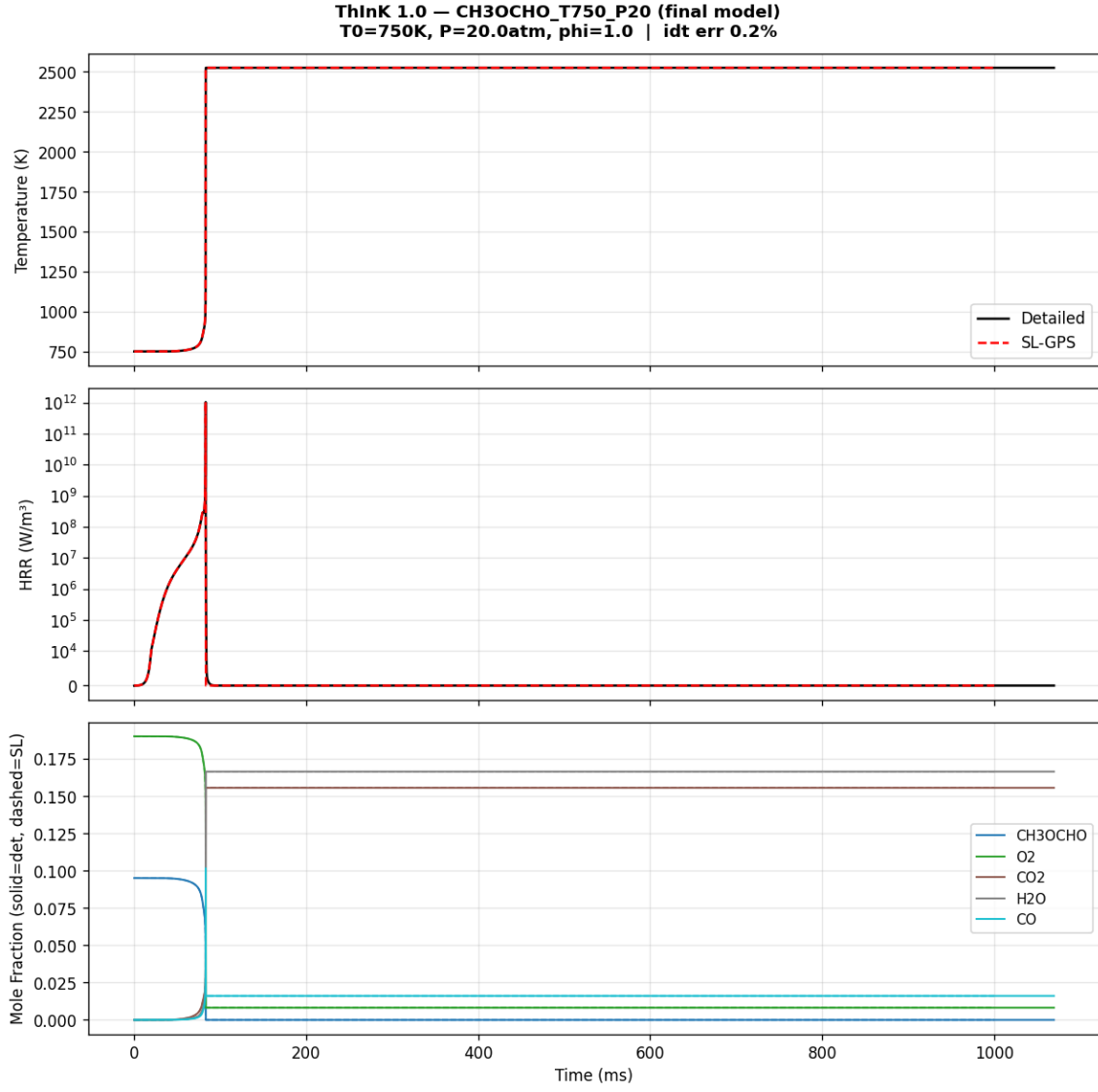


Figure 12: Final model, deep-NTC case T750\_P20 ( $T_0 = 750$  K,  $P = 20$  atm): two-stage ignition reproduced with 0.2% error.

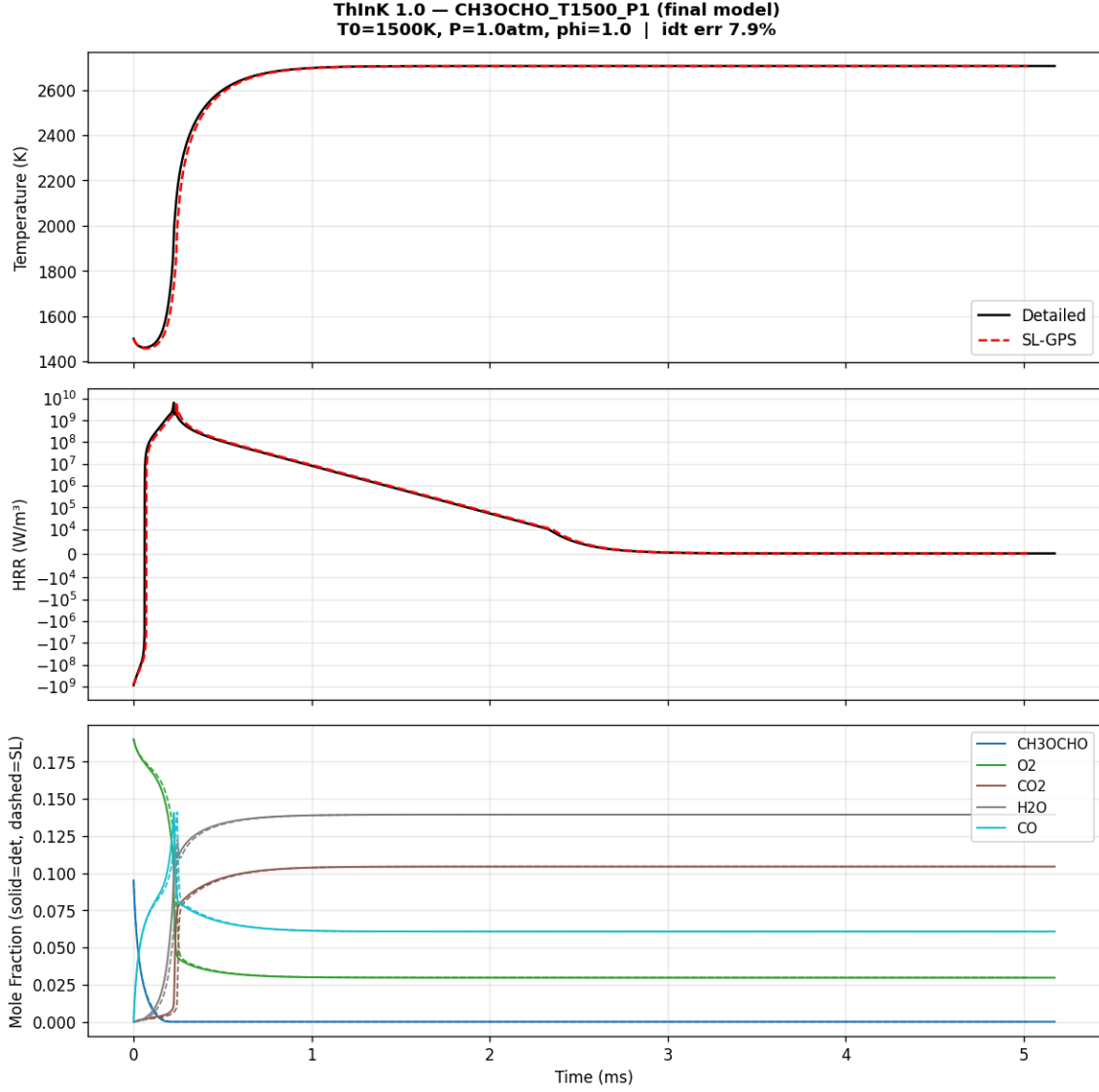


Figure 13: Final model, high-temperature case T1500\_P1 ( $T_0 = 1500$  K,  $P = 1$  atm): the Revision-2 regression (146.7%) is recovered to 7.9% error.

### 5.3 Limitations and Future Work

The idt-scaled timestep rule uses the detailed ignition delay, available here because validation runs the detailed simulation anyway; production use requires a cheap ignition-delay estimate or a state-responsive interval controller. The ANN input ( $T$ ,  $P$ , 14 species) is an information bottleneck at the fresh-mixture state; a hybrid scheme that evaluates classic GPS at the first interval (and optionally at the heat-release-rate transition) and uses the ANN elsewhere would eliminate the residual first-interval prediction error (T1500\_P1, 7.9%) at negligible cost.

### 5.4 Expanded Validation Campaign

To probe the validated envelope further, the test matrix was expanded from 17 to 35 cases: 14 additional in-envelope conditions (low-pressure deep-NTC T750\_P10; mid- $T$ /high- $P$  combinations T950–T1000 at 5–20 atm; rich and lean NTC at 900 K/10 atm; the training-edge cases T1600\_P1 and  $\phi = 1.4$ ), plus four *explicit extrapolation probes* outside the training envelope, prefixed X\_: T700\_P20 (below the 700 K training floor sampling), T900\_P30 (above the 20 atm

pressure ceiling), and  $\phi = 0.5/\phi = 1.6$  at 1100 K/5 atm (beyond the 0.7–1.4 equivalence-ratio range).

The never-list audit was extended to 15 conditions including all extrapolation corners and **passed at every condition**: even outside the training envelope, the ceiling mechanism (always  $\cup$  variable species) reproduces the detailed ignition delay within 0.02%, i.e. the species partition is structurally sound beyond the envelope and any extrapolation error is attributable to ANN prediction alone.

Table 7: Extended never-list audit (data `train_ch3ocho_balanced2`). Blocked = GPS-selected species in the never list; where non-zero, the ceiling mechanism must match detailed ignition within 5%.

| Condition                                     | GPS-required | Blocked | Ceiling idt err | Verdict |
|-----------------------------------------------|--------------|---------|-----------------|---------|
| T1500_P1 / T1200_P1 / T1000_P1                | 38–61        | 0       | —               | ok      |
| T1100_P1_rich / T1600_P1                      | 37–63        | 0       | —               | ok      |
| T900_P10 / T800_P20 / T750_P20                | 87–91        | 14–15   | $\leq 0.02\%$   | ok      |
| T750_P10 / T1000_P10 / T1400_P5_lean          | 70–86        | 5–14    | $\leq 0.01\%$   | ok      |
| X_T700_P20 (extrapolation)                    | 90           | 16      | 0.00%           | ok      |
| X_T900_P30 (extrapolation)                    | 88           | 12      | 0.02%           | ok      |
| X_ $\phi$ 0.5 / X_ $\phi$ 1.6 (extrapolation) | 75–76        | 2–4     | $\leq 0.02\%$   | ok      |

#### 5.4.1 Expanded-Matrix Results

All 31 in-envelope cases pass (worst: 9.8% at the T1600 training-temperature ceiling; all deep-NTC additions at or below 0.7%). Of the four extrapolation probes, three pass — including T700\_P20 *below* the training temperature floor (0.0%) and both equivalence-ratio probes ( $\phi = 0.5$ : 0.4%;  $\phi = 1.6$ : 3.1%). The single failure is the **pressure extrapolation** X\_T900\_P30 (30 atm vs. 20 atm training ceiling): 105.2% error. The model’s validity boundary is therefore set by pressure, consistent with pressure entering the ANN input directly (the input scaler saturates at 20 atm) and with the pressure sensitivity of low-temperature termolecular chemistry. Applications above 20 atm require extending the training pressure range.

Table 8: Expanded validation: 18 additional cases (final model `ch3ocho_balanced2v`, idt-scaled intervals). `X_` cases lie outside the training envelope.

| Case                                    | $T$ (K) | $P$ (atm) | $\phi$ | $\tau_{\text{det}}$ (ms) | $\tau_{\text{SL}}$ (ms) | Error (%) | Pass |
|-----------------------------------------|---------|-----------|--------|--------------------------|-------------------------|-----------|------|
| T750_P10                                | 750     | 10        | 1.0    | 141.05                   | 140.54                  | 0.4       | ✓    |
| T800_P10                                | 800     | 10        | 1.0    | 51.452                   | 51.085                  | 0.7       | ✓    |
| T850_P20                                | 850     | 20        | 1.0    | 10.973                   | 10.903                  | 0.6       | ✓    |
| T950_P5                                 | 950     | 5         | 1.0    | 16.464                   | 16.503                  | 0.2       | ✓    |
| T950_P20                                | 950     | 20        | 1.0    | 3.056                    | 3.037                   | 0.6       | ✓    |
| T1000_P10                               | 1000    | 10        | 1.0    | 3.617                    | 3.595                   | 0.6       | ✓    |
| T1000_P20                               | 1000    | 20        | 1.0    | 1.631                    | 1.721                   | 5.5       | ✓    |
| T1050_P1                                | 1050    | 1         | 1.0    | 18.451                   | 19.408                  | 5.2       | ✓    |
| T900_P10_rich                           | 900     | 10        | 1.3    | 10.869                   | 10.792                  | 0.7       | ✓    |
| T900_P10_lean                           | 900     | 10        | 0.7    | 18.944                   | 18.847                  | 0.5       | ✓    |
| T1300_P10                               | 1300    | 10        | 1.0    | 0.098                    | 0.098                   | 0.2       | ✓    |
| T1400_P1                                | 1400    | 1         | 1.0    | 0.458                    | 0.483                   | 5.4       | ✓    |
| T1200_P5_rich                           | 1200    | 5         | 1.4    | 0.460                    | 0.471                   | 2.3       | ✓    |
| T1600_P1                                | 1600    | 1         | 1.0    | 0.116                    | 0.127                   | 9.8       | ✓    |
| X_T700_P20                              | 700     | 20        | 1.0    | 393.18                   | 393.00                  | 0.0       | ✓    |
| X_T1100_P5_vlean                        | 1100    | 5         | 0.5    | 2.723                    | 2.713                   | 0.4       | ✓    |
| X_T1100_P5_vrich                        | 1100    | 5         | 1.6    | 1.493                    | 1.540                   | 3.1       | ✓    |
| X_T900_P30                              | 900     | 30        | 1.0    | 3.499                    | 7.180                   | 105.2     | ×    |
| <b>Overall (35 cases incl. Table 6)</b> |         |           |        | <b>34/35 passed</b>      |                         |           |      |
| <b>In-envelope (31 cases)</b>           |         |           |        | <b>31/31 passed</b>      |                         |           |      |

The 17 repeated Table 6 cases reproduced their values exactly, confirming run-to-run determinism of the pipeline.

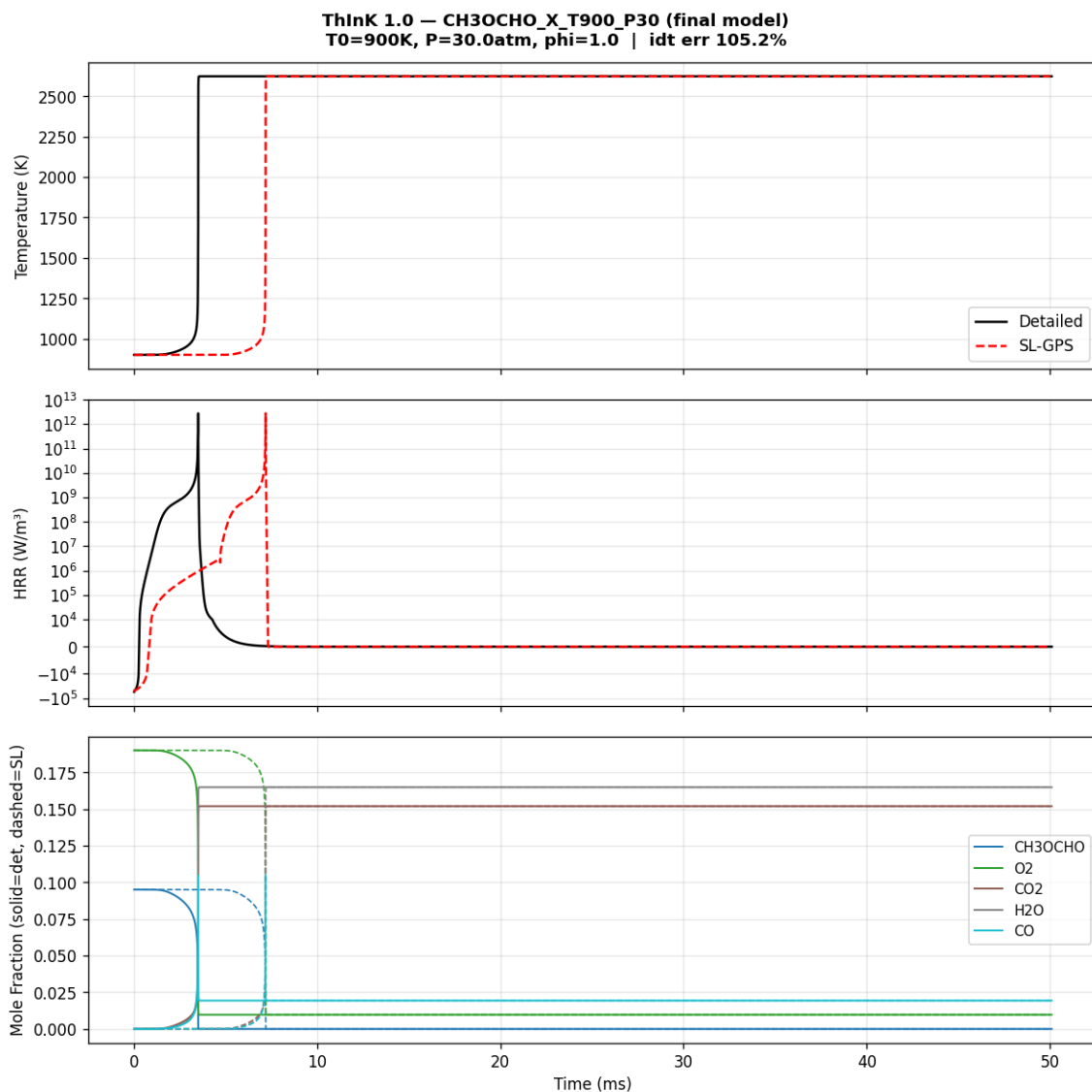


Figure 14: Pressure-extrapolation probe X\_T900\_P30 ( $T_0 = 900$  K,  $P = 30$  atm, outside the 1–20 atm training range): the sole failing case (105.2% ignition-delay error). The input scaler saturates at 20 atm, so the ANN operates on out-of-range input; the never-list audit confirms the species partition itself remains adequate at this condition, isolating the failure to ANN input extrapolation.

## 6 Reproducibility

All scripts are in `validation/ThinkMechanism/`:

```
# Baseline pipeline (data gen + training + validation + plots)
cd validation/ThinkMechanism
python run_pipeline_ch3ocho.py # ~47 min total

# Revision-2 pipeline (Section 6): balanced data, audit, val-loss
# ensemble selection, idt-scaled timesteps, 17-case validation
python make_balanced_data.py train_ch3ocho_balanced2 500 \
    train_ch3ocho train_ch3ocho_expanded train_ch3ocho_hiT
    train_ch3ocho_midT
python audit_never_list.py data/train_ch3ocho_balanced2
python validate_full.py ch3ocho_balanced2v
```

---

## 6.1 Prerequisites

- Cantera  $\geq 3.0.0$  with ThInK 1.0 mechanism (`ThInK_1.0_Mech.yaml`)
- Python 3.10+, TensorFlow 2.x, NumPy, scikit-learn, Matplotlib, Joblib
- $\geq 8$  CPU cores recommended (parallel data generation and ensemble training)

## 6.2 Output Files

Table 9: Pipeline output files.

| Path                                               | Description                                |
|----------------------------------------------------|--------------------------------------------|
| <code>models/ch3ocho/model.h5</code>               | Baseline ANN model                         |
| <code>models/ch3ocho_balanced2v/model.h5</code>    | <b>Final ANN model (v2.1.0)</b>            |
| <code>models/ch3ocho_balanced2v/scaler.pkl</code>  | Final input scaler                         |
| <code>data/train_ch3ocho_balanced2/</code>         | Final balanced training set (3000 samples) |
| <code>data/train_ch3ocho*/</code>                  | Source training batches                    |
| <code>results/ch3ocho/*.png</code>                 | Baseline per-case plots                    |
| <code>results/ch3ocho_final/*.png</code>           | Final-model per-case plots (35 cases)      |
| <code>results_expanded_35case_summary.txt</code>   | 35-case audit + results tables             |
| <code>results/ch3ocho/validation_report.txt</code> | Baseline text report                       |

## References

- [1] ThInK 1.0: A comprehensive thermochemical and kinetic model for oxygenated C<sub>1</sub>–C<sub>3</sub> fuels, *Combustion and Flame*, 2025.
- [2] R. Mishra, A. Nelson, D. Jarrahbashi, “Adaptive global pathway selection using artificial neural networks: A-priori study,” *Combustion and Flame*, vol. 244, p. 112279, 2022.
- [3] X. Gao, S. Yang, W. Sun, “A global pathway selection algorithm for the reduction of detailed chemical kinetic mechanisms,” *Combustion and Flame*, vol. 167, pp. 238–247, 2016.